

Report – Portfolio Management

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Introduction

In this project, we place ourselves in the position of a portfolio manager or an institutional investor with an initial capital to allocate among a set of financial assets. The fundamental question we aim to address is how can we allocate this capital optimally to maximize overall performance while rigorously controlling the level of risk. The set of n available risky assets for investment constitutes our asset universe. A capital allocation is represented by a weight vector $w = (w_1, w_2, \dots, w_n)^T \in \mathbb{R}^n$, where each component w_i corresponds to the proportion of the initial wealth invested in asset i . By definition, this vector must satisfy the self-financing constraint, implying no external capital inflow or withdrawal, which is mathematically expressed as $\sum_{i=1}^n w_i = 1$. If we denote $R = (R_1, R_2, \dots, R_n)^T$ as the vector of random asset returns and $\mu = \mathbb{E}[R]$ as the vector of their expected returns, the expected return of the global portfolio, denoted as μ_p , is defined by the weighted average $\mu_p = w^T \mu = \sum_{i=1}^n w_i \mu_i$.

The universal objective of any investor relies on a permanent trade-off: maximizing the expected return while minimizing the variance, which represents the risk of the allocation. This is precisely the core foundation of Modern Portfolio Theory introduced by Harry Markowitz in 1952. Under this framework, the portfolio's risk is formalized by the variance of its return, given by $\sigma_p^2 = w^T \Sigma w$, where Σ denotes the covariance matrix of the asset returns. Determining the optimal portfolio structure then translates into solving a constrained quadratic optimization problem, which we will solve in the first part.

The Markowitz model assumes by design that the market parameters μ and Σ are perfectly known beforehand. In practice, the random nature of returns forces us to model them. We will initially adopt a standard framework in the second part by assuming that returns follow a Gaussian random vector, entirely defined by its mean vector and its covariance matrix. These parameters will be estimated empirically from historical data. After detailing our methodological choices regarding these calculations in part two, we will also conduct a critical discussion on the relevance of the normality hypothesis against the stylized facts of financial series, such as heavy tails and asymmetry.

Finally, in the third part, we will explain how we achieved a numerical resolution. We will detail the computational approach used to code these optimization algorithms. To make the evaluation of results more concrete, we have developed an interactive graphical animation allowing us to observe and compare in real-time the impact of different allocation strategies on the evolution of the portfolio's net asset value.

1 Part 1: Theoretical Foundations

Modern Portfolio Theory (MPT), introduced by Harry Markowitz in his seminal 1952 paper *Portfolio Selection*, provides a mathematical framework for constructing portfolios that maximize expected return for a given level of risk. The central insight is that diversification across assets reduces portfolio risk, measured by variance, without necessarily sacrificing expected return.

We consider a universe of n assets. The return of asset i is modeled as a random variable X_i , for $1 \leq i \leq n$. Under the Markowitz framework, returns are assumed to be jointly Gaussian, so each asset is fully characterized by its expected return and its covariance with other assets.

The empirical mean return of asset i , estimated over a sample of T observations from t_0 to t_f , is:

$$\mu_i = \frac{1}{T} \sum_{t=t_0}^{t_f} r_{i,t}$$

The empirical covariance matrix $C \in \mathcal{M}_n(\mathbb{R})$, which captures both individual asset variances and pairwise co-movements, is defined by:

$$C_{ij} = \frac{1}{T-1} \sum_{t=t_0}^{t_f} (r_{i,t} - \mu_i)(r_{j,t} - \mu_j)$$

The matrix C is symmetric and positive semi-definite. Its diagonal entries $C_{ii} = \sigma_i^2$ represent the variance of each asset, while off-diagonal entries C_{ij} capture the linear dependence between assets i and j . A portfolio is defined by a weight vector $\mathbf{w} = (w_1, \dots, w_n)^T \in \mathbb{R}^n$, where w_i denotes the fraction of wealth allocated to asset i . The weights satisfy the budget constraint:

$$\mathbf{w}^T \mathbf{1} = 1, \quad \text{where } \mathbf{1} = (1, 1, \dots, 1)^T \in \mathbb{R}^n$$

Under this allocation, the portfolio return is the weighted average of individual asset returns, with expected value $\mathbf{w}^T \boldsymbol{\mu}$ and variance $\mathbf{w}^T C \mathbf{w}$. The investor seeks to minimize portfolio variance for a targeted expected return $\mu^* \in \mathbb{R}$. This yields the following quadratic program:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^T C \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T \mathbf{1} = 1, \\ & \mathbf{w}^T \boldsymbol{\mu} = \mu^*. \end{aligned}$$

This is a convex quadratic optimization problem with linear constraints, which admits a closed-form solution via Lagrange multipliers. By varying the target return μ^* , one traces out the **efficient frontier**: the set of portfolios that achieve the minimum variance for each level of expected return. Rational investors will only hold portfolios lying on this frontier.

To solve the optimization problem, we introduce the Lagrangian with two multipliers $\lambda, \gamma \in \mathbb{R}$:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \mathbf{w}^T C \mathbf{w} - \lambda (\mathbf{w}^T \boldsymbol{\mu} - \mu^*) - \gamma (\mathbf{w}^T \mathbf{1} - 1)$$

Taking the first-order condition with respect to $\mathbf{w}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2C\mathbf{w} - \lambda\boldsymbol{\mu} - \gamma\mathbf{1} = 0$$

Assuming C is invertible (which holds when no asset is a linear combination of others), we obtain:

$$\mathbf{w}^* = \frac{1}{2} C^{-1} (\lambda\boldsymbol{\mu} + \gamma\mathbf{1})$$

The multipliers λ and γ are determined by substituting $\mathbf{w}^*$ back into the two constraints.

Constraint 1: $\mathbf{w}^{*T} \boldsymbol{\mu} = \mu^*$

$$\mathbf{w}^{*T} \boldsymbol{\mu} = \frac{1}{2} (\lambda \boldsymbol{\mu} + \gamma \mathbf{1})^T C^{-1} \boldsymbol{\mu} = \frac{1}{2} (\lambda \boldsymbol{\mu}^T C^{-1} \boldsymbol{\mu} + \gamma \mathbf{1}^T C^{-1} \boldsymbol{\mu}) = \frac{1}{2} (\lambda A + \gamma B) = \mu^*$$

Constraint 2: $\mathbf{w}^{*T} \mathbf{1} = 1$

$$\mathbf{w}^{*T} \mathbf{1} = \frac{1}{2} (\lambda \boldsymbol{\mu} + \gamma \mathbf{1})^T C^{-1} \mathbf{1} = \frac{1}{2} (\lambda \boldsymbol{\mu}^T C^{-1} \mathbf{1} + \gamma \mathbf{1}^T C^{-1} \mathbf{1}) = \frac{1}{2} (\lambda B + \gamma D) = 1$$

We defined the following scalar quantities:

$$A = \boldsymbol{\mu}^T C^{-1} \boldsymbol{\mu}, \quad B = \boldsymbol{\mu}^T C^{-1} \mathbf{1} = \mathbf{1}^T C^{-1} \boldsymbol{\mu}, \quad D = \mathbf{1}^T C^{-1} \mathbf{1}$$

and the determinant $\Delta = AD - B^2 > 0$ (positive by the Cauchy-Schwarz inequality, assuming $\boldsymbol{\mu}$ and $\mathbf{1}$ are not collinear).

Injecting $\mathbf{w}^*$ into $\mathbf{w}^{*T} \boldsymbol{\mu} = \mu^*$ and $\mathbf{w}^{*T} \mathbf{1} = 1$ yields the linear system:

$$\frac{1}{2} \begin{pmatrix} A & B \\ B & D \end{pmatrix} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} \mu^* \\ 1 \end{pmatrix}$$

Inverting this 2×2 system:

$$\lambda = \frac{2(D\mu^* - B)}{\Delta}, \quad \gamma = \frac{2(A - B\mu^*)}{\Delta}$$

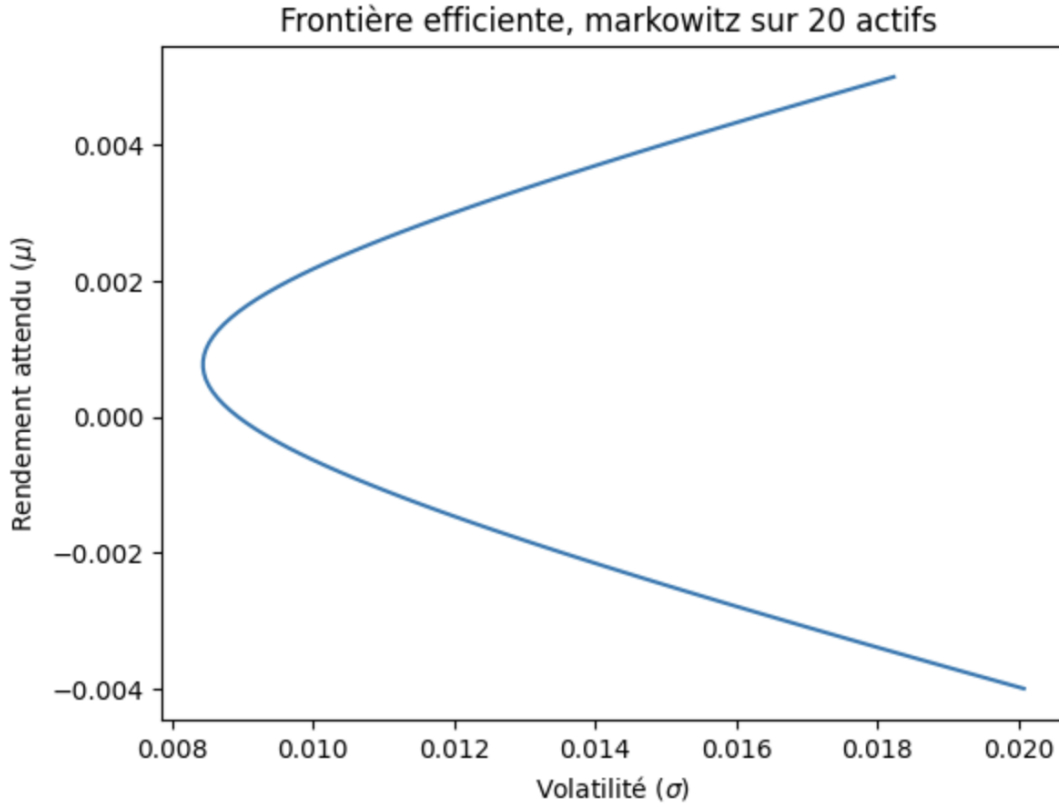


Figure 1: Tracé de la frontière efficiente pour les 20 actifs

Substituting λ and γ back into the expression for $\mathbf{w}^*$:

$$\mathbf{w}^* = \frac{D\mu^* - B}{\Delta} C^{-1} \boldsymbol{\mu} + \frac{A - B\mu^*}{\Delta} C^{-1} \mathbf{1}$$

This can be rewritten as a convex combination of two reference portfolios:

$$\mathbf{w}^* = \alpha \mathbf{w}_\mu + (1 - \alpha) \mathbf{w}_1$$

where:

$$\mathbf{w}_\mu = \frac{C^{-1} \boldsymbol{\mu}}{\mathbf{1}^T C^{-1} \boldsymbol{\mu}}, \quad \mathbf{w}_1 = \frac{C^{-1} \mathbf{1}}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad \alpha = \frac{D\mu^* - B}{\Delta} \cdot B$$

This is the **two-fund separation theorem**: any efficient portfolio is a linear combination of just two fundamental portfolios, regardless of the target return μ^* . By symmetry of the optimization problem, one can instead fix the portfolio variance $\sigma^{*2} = \mathbf{w}^T C \mathbf{w}$ and seek the allocation that maximizes the expected return $\mathbf{w}^T \boldsymbol{\mu}$:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mathbf{w}^T \boldsymbol{\mu} \\ \text{subject to} \quad & \mathbf{w}^T C \mathbf{w} = \sigma^{*2}, \\ & \mathbf{w}^T \mathbf{1} = 1. \end{aligned}$$

The Lagrangian, with multiplier η on the (now nonlinear) variance constraint and γ on the budget constraint, is:

$$\mathcal{L}(\mathbf{w}, \eta, \gamma) = \mathbf{w}^T \boldsymbol{\mu} - \eta (\mathbf{w}^T C \mathbf{w} - \sigma^{*2}) - \gamma (\mathbf{w}^T \mathbf{1} - 1)$$

Setting the gradient with respect to $\mathbf{w}$ to zero:

$$\boldsymbol{\mu} - 2\eta C \mathbf{w} - \gamma \mathbf{1} = 0 \implies \mathbf{w}^* = \frac{1}{2\eta} C^{-1} (\boldsymbol{\mu} - \gamma \mathbf{1})$$

This expression has exactly the same structure as in the variance-minimization problem (it is again a linear combination of $C^{-1} \boldsymbol{\mu}$ and $C^{-1} \mathbf{1}$), which confirms that both formulations trace out the **same efficient frontier**: minimizing variance for a given return and maximizing return for a given variance are two dual descriptions of the same set of optimal portfolios. In practice, η and γ are pinned down by substituting $\mathbf{w}^*$ back into the budget and variance constraints, exactly as in the previous derivation, leading to the same expression of $\mu^*(\sigma^{*2})$ obtained by inverting the hyperbola $\sigma^2(\mu^*) = \frac{D\mu^{*2} - 2B\mu^* + A}{\Delta}$.

However, investors are sometimes prevented, by mandate or regulation, from taking short positions, that is, selling a stock they do not currently own. This corresponds to forbidding negative coordinates in $\mathbf{w}$, and the long-only constraint is imposed as:

$$w_i \geq 0, \quad \forall i \in \{1, \dots, n\}.$$

Complete Solution with the Long-Only Constraint via KKT

We solve explicitly:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^T C \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T \mathbf{1} = 1, \\ & \mathbf{w}^T \boldsymbol{\mu} = \mu^*, \\ & w_i \geq 0, \quad \forall i \in \{1, \dots, n\}. \end{aligned}$$

Lagrangian. Introduce multipliers $\lambda, \gamma \in \mathbb{R}$ for the equality constraints and $\nu_i \geq 0$ for each inequality constraint $-w_i \leq 0$:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma, \boldsymbol{\nu}) = \mathbf{w}^T C \mathbf{w} - \lambda (\mathbf{w}^T \boldsymbol{\mu} - \mu^*) - \gamma (\mathbf{w}^T \mathbf{1} - 1) - \boldsymbol{\nu}^T \mathbf{w}$$

KKT conditions. Since the objective is convex and the constraints are affine, the KKT conditions are necessary and sufficient for global optimality:

$$\begin{aligned}
& \text{(stationarity)} & 2C\mathbf{w}^* - \lambda\boldsymbol{\mu} - \gamma\mathbf{1} - \boldsymbol{\nu} &= 0 \\
& \text{(primal feasibility)} & \mathbf{w}^{*T}\mathbf{1} = 1, \quad \mathbf{w}^{*T}\boldsymbol{\mu} = \mu^*, \quad w_i^* \geq 0 \quad \forall i \\
& \text{(dual feasibility)} & \nu_i \geq 0 \quad \forall i \\
& \text{(complementary slackness)} & \nu_i w_i^* = 0 \quad \forall i
\end{aligned}$$

Partition of assets. Let $\mathcal{I} = \{i : w_i^* > 0\}$ be the set of active (held) assets and $\mathcal{J} = \{i : w_i^* = 0\}$ the set of excluded assets, with $\mathcal{I} \cup \mathcal{J} = \{1, \dots, n\}$.

By complementary slackness, $\nu_i = 0$ for all $i \in \mathcal{I}$. Restricting the stationarity condition to the active set, and writing $C_{\mathcal{I}}$, $\boldsymbol{\mu}_{\mathcal{I}}$, $\mathbf{1}_{\mathcal{I}}$ for the submatrix and subvectors restricted to $\mathcal{I}$:

$$2C_{\mathcal{I}}\mathbf{w}_{\mathcal{I}}^* = \lambda\boldsymbol{\mu}_{\mathcal{I}} + \gamma\mathbf{1}_{\mathcal{I}}$$

This is exactly the unconstrained stationarity equation, but restricted to the sub-universe $\mathcal{I}$. Assuming $C_{\mathcal{I}}$ is invertible:

$$\boxed{\mathbf{w}_{\mathcal{I}}^* = \frac{1}{2}C_{\mathcal{I}}^{-1}(\lambda\boldsymbol{\mu}_{\mathcal{I}} + \gamma\mathbf{1}_{\mathcal{I}}), \quad w_j^* = 0 \quad \forall j \in \mathcal{J}}$$

with λ, γ pinned down, exactly as before, by substituting into the two equality constraints restricted to $\mathcal{I}$:

$$\begin{aligned}
A_{\mathcal{I}} &= \boldsymbol{\mu}_{\mathcal{I}}^T C_{\mathcal{I}}^{-1} \boldsymbol{\mu}_{\mathcal{I}}, & B_{\mathcal{I}} &= \boldsymbol{\mu}_{\mathcal{I}}^T C_{\mathcal{I}}^{-1} \mathbf{1}_{\mathcal{I}}, & D_{\mathcal{I}} &= \mathbf{1}_{\mathcal{I}}^T C_{\mathcal{I}}^{-1} \mathbf{1}_{\mathcal{I}} \\
\lambda &= \frac{2(D_{\mathcal{I}}\mu^* - B_{\mathcal{I}})}{\Delta_{\mathcal{I}}}, & \gamma &= \frac{2(A_{\mathcal{I}} - B_{\mathcal{I}}\mu^*)}{\Delta_{\mathcal{I}}}, & \Delta_{\mathcal{I}} &= A_{\mathcal{I}}D_{\mathcal{I}} - B_{\mathcal{I}}^2
\end{aligned}$$

Validity check. This candidate solution is optimal if and only if two conditions hold simultaneously:

$$w_i^* > 0 \quad \forall i \in \mathcal{I}, \quad \nu_j = -(2(C\mathbf{w}^*)_j - \lambda\mu_j - \gamma) \geq 0 \quad \forall j \in \mathcal{J}$$

The first condition checks that the held assets indeed have positive weight; the second checks that excluding the remaining assets is consistent with the optimality conditions (their "implicit return" is dominated).

Resolution method. The KKT analysis above is the *theoretical* characterization of the long-only optimum: it shows that the solution is the unconstrained two-fund formula restricted to an a priori unknown active set $\mathcal{I}$. Conceptually, this active set can be identified by the classical *active-set algorithm* for quadratic programming: start with $\mathcal{I} = \{1, \dots, n\}$, compute $\mathbf{w}^*$ via the unconstrained formula, and if some weights are negative, remove the most negative one(s) and repeat on the reduced set until all retained weights are non-negative.

In our implementation, however, we do not run this combinatorial loop explicitly. The closed-form Lagrangian solution is used only for the *equality-constrained* case (budget and target return), while the long-only problem, and more generally any problem with weight caps $w_i \leq w_{\max}$ or transaction-cost penalties, is handed directly to a general convex quadratic-programming solver (CVXPY with the CLARABEL backend, see Part 3). Since the objective is convex and all constraints are affine, this solver returns the exact global optimum characterized by the KKT conditions above, without us having to enumerate the active set by hand. The active-set derivation therefore explains *why* the long-only frontier is piecewise-quadratic (it is the two-fund formula applied recursively on shrinking subsets of assets), while the convex solver is what actually computes it in practice.

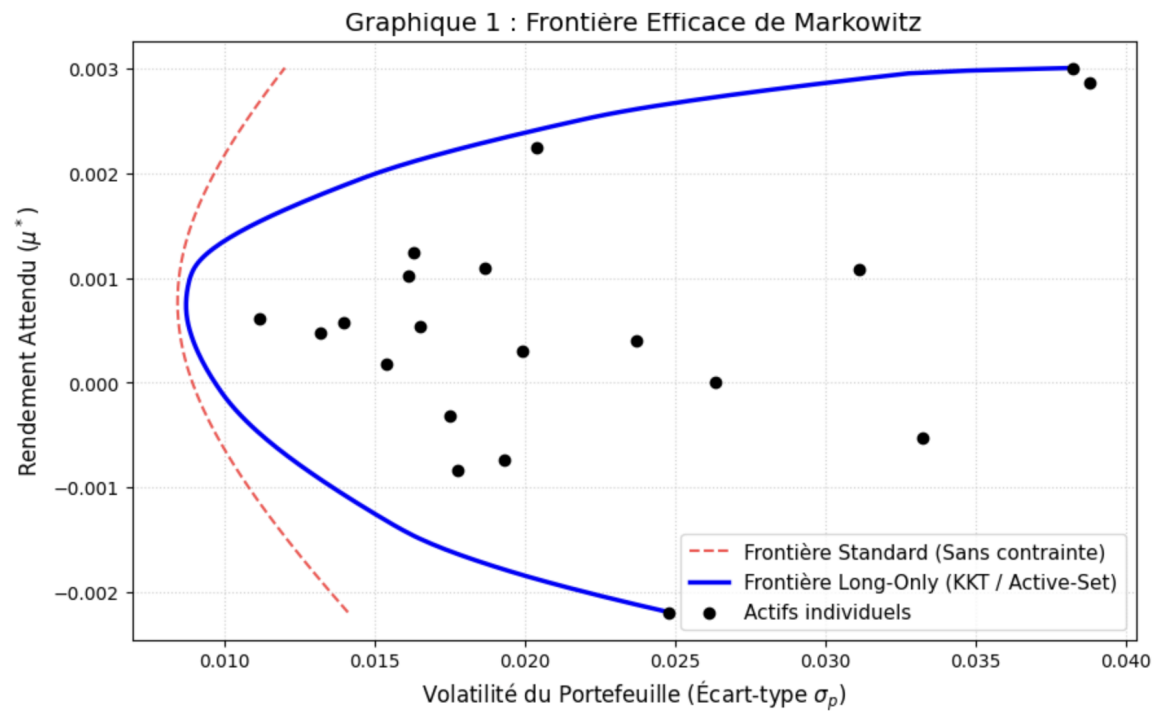


Figure 2: Frontière efficace avec long-only comparé a la version precedente

2 Part 2: Probability and statistics

As stated in the introduction, the Markowitz solution is optimal if and only if asset returns are Gaussian and the mean vector $\boldsymbol{\mu}$ and the covariance matrix C are known exactly. In practice, however, these parameters are never directly observable: an investor only has access to a finite history of past returns, from which $\boldsymbol{\mu}$ and C must be estimated statistically. This estimation step is a critical, and often underappreciated, source of error in the mean-variance framework, since the entire optimization is built on top of these estimated parameters rather than the true ones. We consider a universe of n assets. The return of asset i is modeled as a random variable X_i ,

for $1 \leq i \leq n$. As a first and simplest approach, returns are assumed to be jointly Gaussian and independently and identically distributed (i.i.d.) over time, so each asset is fully characterized by its expected return and its covariance with the other assets, both assumed constant throughout the estimation window. The most natural estimators are the sample mean and the sample covariance, computed from a history of T observations between t_0 and t_f . The empirical mean return of asset i is:

$$\mu_i = \frac{1}{T} \sum_{t=t_0}^{t_f} r_{i,t}$$

By the law of large numbers, this estimator is unbiased and converges to the true mean $\mathbb{E}[X_i]$ as $T \rightarrow \infty$, but it remains a noisy estimate over the short and moderate sample sizes typically available in practice (a few years of daily or monthly returns). The empirical covariance matrix $C \in \mathcal{M}_n(\mathbb{R})$, which captures both the individual variance of each asset and the pairwise co-movements between assets, is defined by:

$$C_{ij} = \frac{1}{T-1} \sum_{t=t_0}^{t_f} (r_{i,t} - \mu_i)(r_{j,t} - \mu_j)$$

The matrix C is symmetric and positive semi-definite by construction. Its diagonal entries $C_{ii} = \sigma_i^2$ represent the variance of each individual asset, while its off-diagonal entries C_{ij} capture the linear dependence between assets i and j , normalized as the covariance $C_{ij} = \rho_{ij}\sigma_i\sigma_j$, where ρ_{ij} is the correlation coefficient between assets i and j .

While straightforward to implement, this naive sample-based approach suffers from several well-documented limitations, which directly affect the reliability of the resulting Markowitz allocation:

- **Estimation error on $\boldsymbol{\mu}$.** The sample mean is known to be a particularly noisy estimator of expected returns: its standard error scales as $\sigma_i/\sqrt{T}$, which decreases only slowly with the sample size. Since portfolio weights in the Markowitz solution depend linearly on $C^{-1}\boldsymbol{\mu}$, small estimation errors in $\boldsymbol{\mu}$ can translate into large and unstable swings in the optimal allocation, a phenomenon often referred to as *error maximization*.
- **Ill-conditioning of C .** When the number of assets n is large relative to the number of observations T (a common situation when n approaches or exceeds T), the sample covariance matrix becomes ill-conditioned or even singular, making its inversion numerically unstable and the resulting weights extremely sensitive to small perturbations in the data.
- **Non-stationarity.** The i.i.d. assumption implies that $\boldsymbol{\mu}$ and C are constant over the estimation window, which is rarely true in practice: volatility tends to cluster over time, and correlations between assets are known to shift, particularly during periods of market stress, when diversification benefits tend to disappear precisely when they are most needed.

These limitations motivate the use of more robust estimation techniques.

To address the non-stationarity limitation of the naive estimator, the **Exponentially Weighted Moving Average (EWMA)** method assigns exponentially decreasing weights to past observations, rather than weighting all T historical returns equally. Recent observations, more informative about the current market regime, thus contribute more to the estimate than distant ones. The weight assigned to the observation at time t , within a window ending at T , is:

$$w_t = (1 - \lambda) \lambda^{T-1-t}$$

where $\lambda \in (0, 1)$ is the decay factor controlling how quickly the influence of past observations fades. The corresponding covariance estimator is:

$$\hat{\Sigma} = X_w^T X_w$$

where X_w is the matrix of returns weighted by $\sqrt{w_t}$, i.e. each return row $\mathbf{r}_t$ is scaled by $\sqrt{w_t}$ before computing the cross products. Following the standard RiskMetrics convention, we do *not* subtract a sample mean before forming these products: at the daily horizon the mean return is negligible relative to volatility ($|\mu_i| \ll \sigma_i$), so the estimator effectively targets the (non-centered) second moment $E[\mathbf{r}\mathbf{r}^T]$, which coincides with the covariance up to a $\mathcal{O}(\mu\mu^T)$ term. Writing $\hat{\Sigma}$ as a Gram matrix $X_w^T X_w$ also guarantees that it remains positive semi-definite by construction, an important property since the Markowitz solution requires inverting (or factorizing) this matrix. The decay factor λ directly controls the trade-off between responsiveness and stability: a value close to 1 behaves similarly to the naive equal-weighted estimator, while a lower value makes the estimate highly reactive to recent shocks, at the cost of higher estimation noise. The RiskMetrics methodology, widely used in practice, recommends:

$$\lambda = 0.94 \text{ (daily data),} \quad \lambda = 0.97 \text{ (monthly data)}$$

The practical interpretation of λ is best captured by the *effective half-life* of the weighting scheme, i.e. the number of periods after which an observation's weight has been reduced by half:

$$\text{half-life} = \frac{\ln 2}{\ln(1/\lambda)} \approx 11 \text{ days } (\lambda = 0.94)$$

This makes EWMA particularly well suited to capturing volatility clustering and shifting correlations, directly addressing the non-stationarity concern raised above, at the cost of introducing an additional hyperparameter λ that must itself be calibrated or chosen by convention.

To address the ill-conditioning problem, which arises when the number of assets n is large relative to the number of observations T , the **Ledoit-Wolf shrinkage** estimator regularizes the sample covariance matrix by pulling it towards a more stable, structured target matrix F (for instance, a diagonal matrix, or a matrix implied by a single-factor or constant-correlation model). The shrinkage estimator takes the form:

$$\hat{\Sigma}_{\text{shrink}} = \delta F + (1 - \delta) C$$

where C is the sample covariance matrix defined earlier, and $\delta \in [0, 1]$ is the shrinkage intensity, optimally chosen by Ledoit and Wolf to minimize the expected quadratic loss between $\hat{\Sigma}_{\text{shrink}}$ and the true unknown covariance matrix. This approach deliberately introduces a small bias toward the target F , in exchange for a substantial reduction in the variance of the estimator, an instance of the classical **bias-variance trade-off**. The resulting matrix is better conditioned than the raw sample covariance, its smallest eigenvalues are pulled upward and its largest eigenvalues are pulled downward, which makes its inversion numerically stable even when n approaches or exceeds T , a regime in which the naive sample covariance matrix is singular or near-singular. In the context of Markowitz optimization, this directly mitigates the error maximization problem: since the optimal weights depend on C^{-1} , a better conditioned and more stable covariance estimate translates into less extreme and more robust portfolio allocations, even though the estimator itself is no longer unbiased.

EWMA and Ledoit-Wolf shrinkage address two distinct, complementary weaknesses of the naive empirical estimator: the former tackles the assumption of time-invariant parameters by adapting to recent market conditions, while the latter tackles the curse of dimensionality and numerical instability by regularizing the covariance structure itself. In practice, the two approaches are often combined, for instance by applying Ledoit-Wolf shrinkage to an EWMA-weighted covariance matrix, in order to obtain an estimator that is both responsive to changing market regimes and numerically well behaved.

In the following figure, we plot the correlation matrices obtained from the empirical (sample) covariance, a rolling window of 30 days, the EWMA method, and the Ledoit-Wolf method. κ denotes the condition number of the covariance matrix. A value of κ close to 1 indicates that the matrix is well-conditioned, whereas a high value indicates that it is poorly conditioned.

We use the daily returns of 8 stocks between 2024 and 2026.

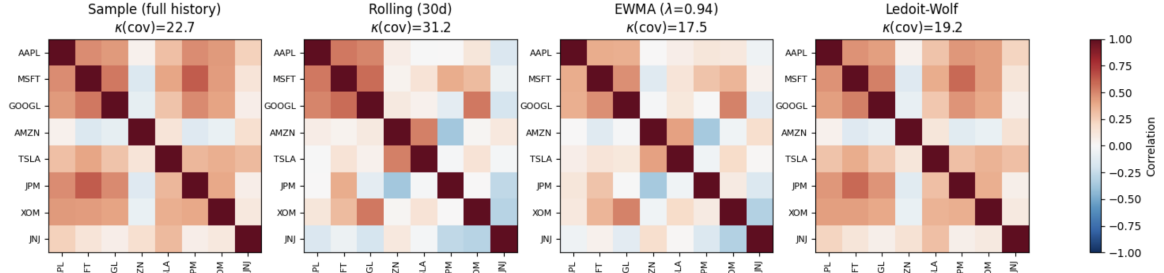


Figure 3: The comparison of the 4 correlation matrices

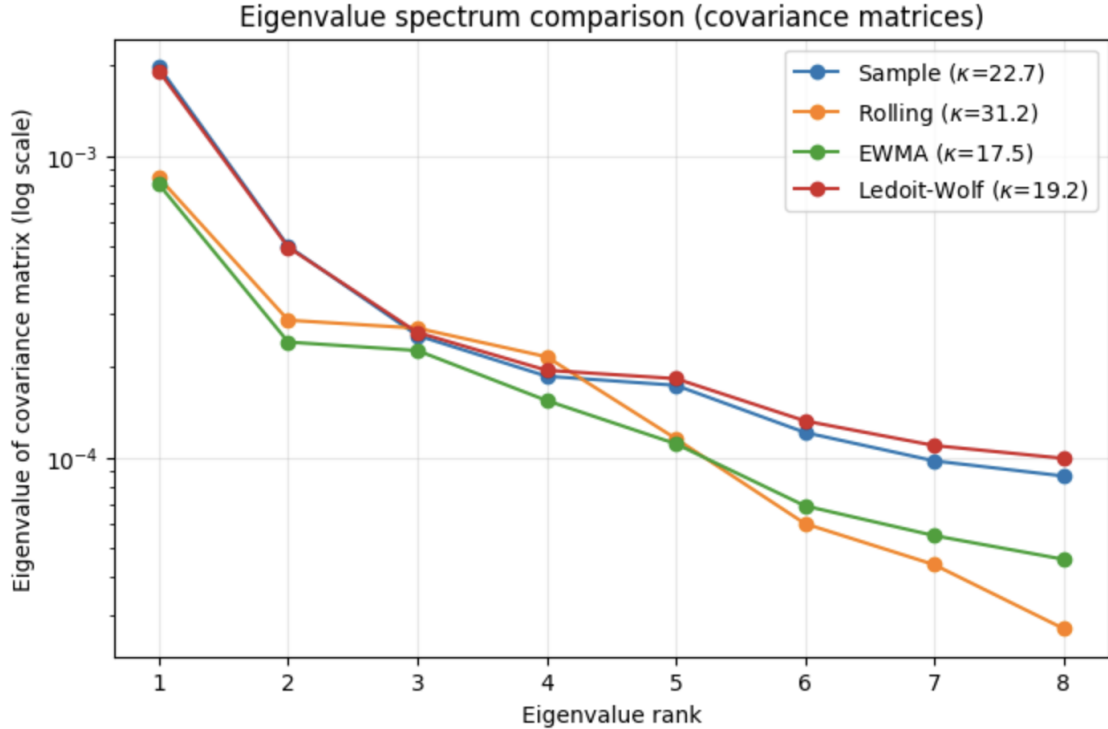


Figure 4: The comparison of the eigenvalues

As we did for the covariance matrix we can also improve the estimation of returns. The sample mean $\hat{\boldsymbol{\mu}} = \frac{1}{T} \sum_t \mathbf{r}_t$ is an unbiased estimator of the true mean vector $\boldsymbol{\mu}$, but it is very noisy in practice, particularly when the number of assets n is large relative to the number of observations T . A natural solution is to shrink the sample mean toward a common target, exactly as Ledoit-Wolf shrinks the covariance matrix. The **James-Stein estimator** does precisely this: instead of trusting each asset's sample mean individually, it pulls all estimates toward a common reference value μ_0 , typically chosen as the grand mean of all assets:

$$\mu_0 = \frac{1}{n} \sum_{i=1}^n \hat{\mu}_i$$

The estimator then takes the form:

$$\hat{\mu}_i^{\text{JS}} = (1 - \delta) \hat{\mu}_i + \delta \mu_0$$

where $\delta \in [0, 1]$ is the shrinkage intensity, which controls how strongly each individual estimate is pulled toward the common mean μ_0 . When $\delta = 0$ we recover the raw sample mean; when $\delta = 1$ all assets are assigned the same expected return μ_0 .

Why does this work? The intuition is straightforward: in a universe of n assets, the sample mean of each asset is estimated from only T observations and therefore contains a large amount of noise. Some assets will have an artificially high sample mean simply by chance, and others an artificially low one. By shrinking all estimates toward a common value, the James-Stein estimator reduces these extreme, noise-driven deviations, at the cost of introducing a small bias. The key theoretical result, known as *Stein's phenomenon*, guarantees that for $n \geq 3$ this trade-off is always favorable: the James-Stein estimator has a strictly lower mean squared error than the sample mean, for any true value of $\boldsymbol{\mu}$:

$$\mathbb{E} [\|\hat{\boldsymbol{\mu}}^{\text{JS}} - \boldsymbol{\mu}\|^2] < \mathbb{E} [\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2]$$

In the portfolio context, this directly translates into more stable and less extreme optimal weights $\mathbf{w}^* \propto C^{-1} \boldsymbol{\mu}$, since the input $\boldsymbol{\mu}$ fed into the optimizer is less contaminated by estimation noise. The James-Stein estimator is therefore the natural complement to Ledoit-Wolf shrinkage: the former regularizes the mean vector $\boldsymbol{\mu}$, the latter regularizes the covariance matrix C , and the two are typically combined in practice to obtain a fully robust Markowitz optimizer.

Limitations of the Gaussian Assumption: Higher Moments and Risk Indicators

The entire framework developed above relies on the assumption that asset returns are Gaussian, fully characterized by their first two moments, the mean μ_i and the variance σ_i^2 . In practice, however, financial returns frequently exhibit departures from normality, in particular fat tails (extreme events occurring more often than a Gaussian model would predict) and asymmetry between gains and losses. Several indicators are used to detect and quantify these deviations.

Skewness measures the asymmetry of the return distribution around its mean. It is estimated empirically as the standardized third central moment:

$$\text{Skew}_i = \frac{\frac{1}{T} \sum_{t=t_0}^{t_f} (r_{i,t} - \mu_i)^3}{\sigma_i^3}$$

A negative skewness indicates a distribution with a longer left tail, large losses are rare but more extreme than large gains, a pattern commonly observed in equity returns and particularly undesirable for risk-averse investors.

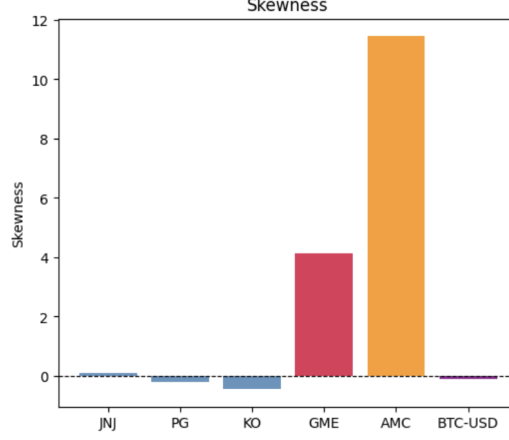


Figure 5: The skewness for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are highly non-gaussian

Kurtosis measures the thickness of the tails relative to a Gaussian distribution. It is estimated as the standardized fourth central moment:

$$\text{Kurt}_i = \frac{\frac{1}{T} \sum_{t=t_0}^{t_f} (r_{i,t} - \mu_i)^4}{\sigma_i^4}$$

A Gaussian distribution has a kurtosis of exactly 3. Financial returns typically display *excess kurtosis* ($\text{Kurt} > 3$, also called leptokurtosis), meaning extreme returns, both positive and negative, occur far more frequently than the Gaussian model predicts. This is usually summarized by the excess kurtosis $\text{Kurt}_i - 3$.

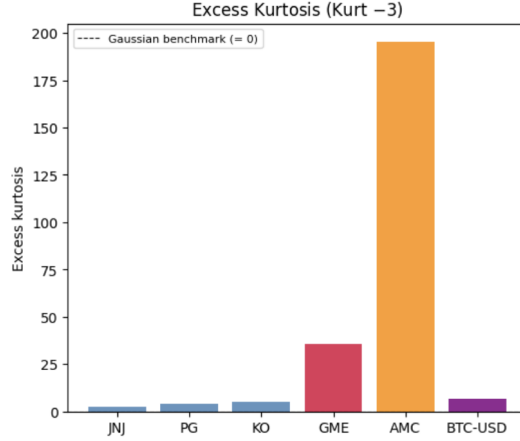


Figure 6: The leptokurtosis coefficient for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are highly non-gaussian

The Gini coefficient, more commonly used to measure inequality in income distributions, can also be applied to returns as a measure of dispersion alternative to variance, less sensitive to the Gaussian assumption. It is defined from the mean absolute difference between all pairs of observations:

$$\text{Gini}_i = \frac{1}{2T^2\mu_i} \sum_{s=t_0}^{t_f} \sum_{t=t_0}^{t_f} |r_{i,s} - r_{i,t}|$$

A higher Gini coefficient indicates a more dispersed and unequal distribution of returns, capturing extreme inequality between observations in a way variance alone does not.

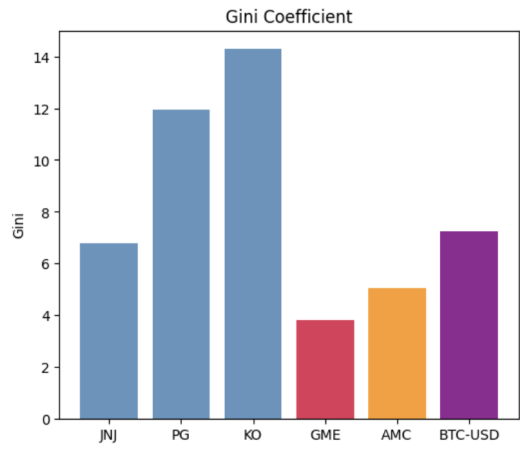


Figure 7: The Gini coefficient for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are absolutely non-gaussian

Taken together, these indicators allow the analyst to flag assets whose return distribution departs significantly from normality, which directly threatens the validity of the mean-variance framework: when skewness and excess kurtosis are large, variance no longer fully captures the risk faced by the investor, since it ignores the asymmetry and the probability of extreme losses.

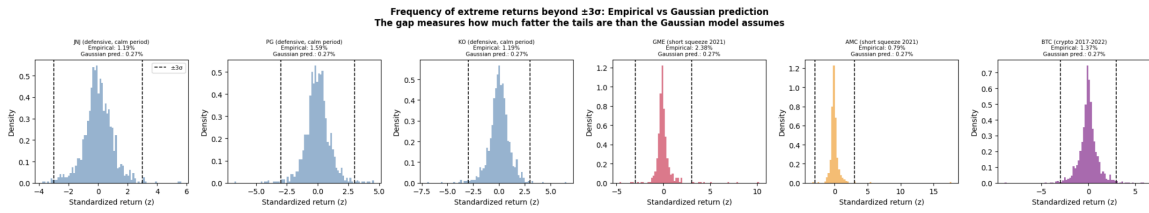


Figure 8: The distribution of return for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are absolutely non-gaussian

Value at Risk as a Complementary Risk Measure

To address this limitation, the **Value at Risk (VaR)** is commonly introduced as a risk measure that directly targets the tail of the distribution rather than its overall dispersion. For a confidence level α (typically 95% or 99%), the VaR of a portfolio over a holding period is defined as the loss threshold that will not be exceeded with probability α :

$$\mathbb{P}(R_p \leq -\text{VaR}_\alpha) = 1 - \alpha$$

where $R_p = \mathbf{w}^T \mathbf{r}$ is the portfolio return. Under the Gaussian assumption, VaR_α has a simple closed form, $\text{VaR}_\alpha = -\mathbf{w}^T \boldsymbol{\mu} + z_\alpha \sqrt{\mathbf{w}^T \mathbf{C} \mathbf{w}}$ where z_α is the corresponding quantile of the standard normal distribution. However, when returns exhibit fat tails or skewness, this Gaussian VaR systematically underestimates the true risk of large losses, and the VaR is instead estimated empirically (historical simulation) or via Monte Carlo simulation from the actual, non-Gaussian distribution of returns.

In portfolio allocation, VaR is used in two main ways. First, as a constraint added to the classical mean-variance problem, the investor minimizes variance (or maximizes return) subject to the additional condition that the portfolio's VaR does not exceed a maximum acceptable threshold VaR^* , which prevents the optimizer from

selecting portfolios that look efficient in mean-variance terms but carry a disproportionate risk of extreme loss. Second, as an alternative objective function, replacing the variance entirely: the investor minimizes $\text{VaR}_\alpha(\mathbf{w})$ directly, subject to the budget and return constraints, which better accounts for tail risk but generally loses the closed-form tractability of the Markowitz solution, since VaR is not, in general, a convex function of $\mathbf{w}$ (a limitation that motivates the use of the related but convex Conditional Value at Risk, or Expected Shortfall, in more advanced allocation frameworks).

Monte Carlo Simulation for Efficient Frontier An alternative to solving the quadratic program analytically is to approximate the efficient frontier numerically using a Monte Carlo approach. The method consists of randomly generating a large number of portfolios, computing their expected return and variance, and retaining the ones that dominate the others.

Procedure.

1. Generate N random weight vectors $\mathbf{w}^{(k)} \in \mathbb{R}^n$, for $k = 1, \dots, N$, typically by drawing n uniform or Dirichlet-distributed random numbers and normalizing so that $\mathbf{w}^{(k)T} \mathbf{1} = 1$ (the Dirichlet distribution is preferred as it also naturally enforces $w_i \geq 0$, making it convenient for simulating the long-only case).
2. For each portfolio, compute its expected return $\mu^{(k)} = \mathbf{w}^{(k)T} \boldsymbol{\mu}$ and variance $\sigma^{(k)2} = \mathbf{w}^{(k)T} \mathbf{C} \mathbf{w}^{(k)}$.
3. Plot all N portfolios in the (σ, μ) plane: the resulting point cloud fills the feasible region.
4. Identify the upper-left boundary of the cloud, i.e. for each level of risk, the portfolio achieving the highest return (or equivalently, for each return level, the lowest risk). This empirical boundary approximates the efficient frontier.

Accuracy of the method: The Monte Carlo approach is intuitive and requires no matrix inversion, but it only provides an *approximation* of the true frontier, and from the inside: since the sampled portfolios are random draws from the feasible set, the boundary of the cloud will always lie strictly below (in return) or to the right (in risk) of the true analytical frontier derived in the previous sections. The gap between the simulated boundary and the true frontier depends on:

- **The number of simulations N .** As $N \rightarrow \infty$, the point cloud becomes dense enough that its boundary converges to the true frontier, but convergence is slow: tracing the frontier accurately in a universe of n assets requires sampling a region of an $(n-1)$ -dimensional simplex, and the probability of randomly drawing a portfolio close to the true optimum decreases sharply as n increases. In practice, a few thousand portfolios suffice for $n \leq 5$ assets, while tens or hundreds of thousands are needed beyond $n = 10$ to obtain a visually smooth and accurate boundary.
- **The sampling distribution.** A naive uniform sampling on the simplex tends to concentrate samples near the center (equally weighted portfolios), under-sampling the extreme, concentrated portfolios that often lie on the frontier itself, especially near its endpoints (minimum variance and maximum return portfolios). This further degrades precision unless the sampling scheme is adapted.

For this reason, Monte Carlo simulation is mainly used as a **visualization and intuition-building tool** to illustrate the shape of the feasible set and the position of the efficient frontier as its boundary, rather than as a precise computational method. For an accurate or production-grade frontier, the closed-form solution (or its constrained quadratic programming counterpart) derived earlier remains strictly superior, since it locates the exact frontier without sampling error and at a negligible computational cost.

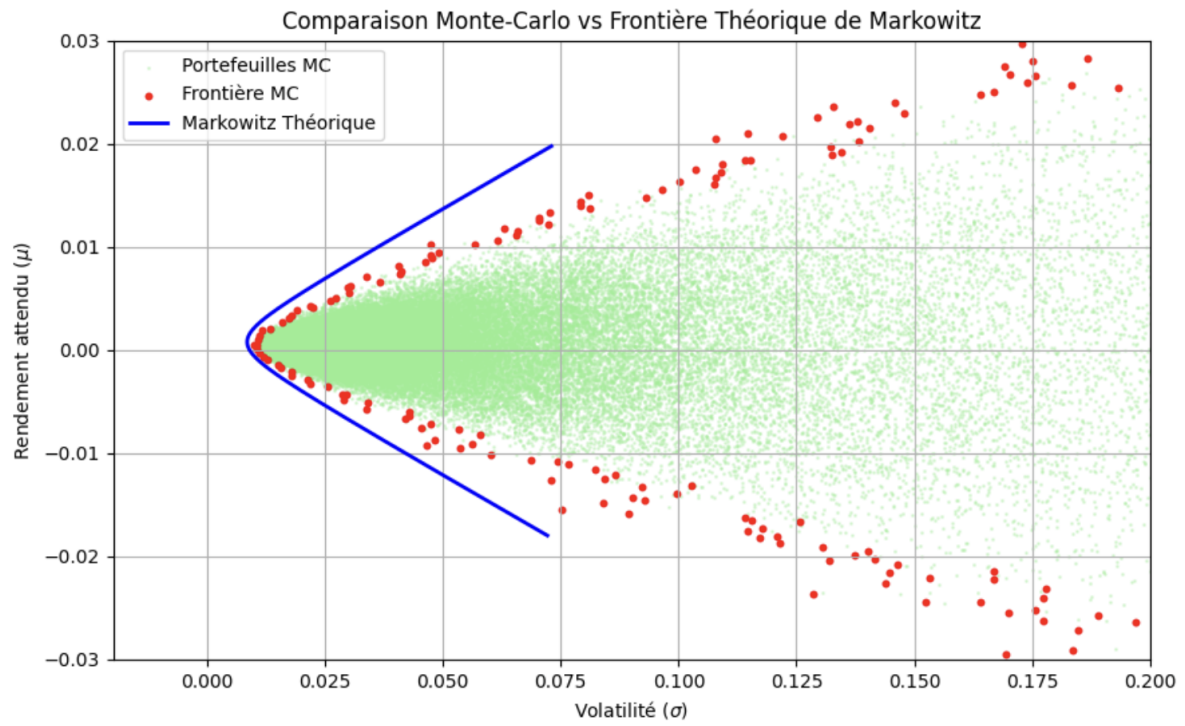


Figure 9: Plot of the theoretical efficient frontier and Monte Carlo simulated portfolios under Gaussian assumption

3 Part 3: Numerical Resolution and Representation

All numerical experiments presented in this report were implemented in Python. Historical price data was retrieved using the `yfinance` library, which provides access to Yahoo Finance market data. For each asset in the universe, we collected daily adjusted closing prices over the chosen estimation window, using the `yf.download()` function. Daily *simple* (arithmetic) returns were then computed from the closing price series using the `pct_change()` method, which computes for each day t :

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

We deliberately use simple returns rather than logarithmic returns, because the portfolio return aggregates *linearly across assets* as $r_{p,t} = \mathbf{w}^T \mathbf{r}_t = \sum_i w_i r_{i,t}$, which is exactly the quantity the mean-variance framework operates on; log returns, by contrast, aggregate linearly across *time* but not across assets, so $\mathbf{w}^T$ of log returns is not the log return of the portfolio. Missing values introduced by market closures or data gaps were removed using `dropna()`, ensuring that the return matrix fed into the optimizer is complete and well-defined.

Optimization method

From the cleaned return matrix, we estimated the mean vector $\boldsymbol{\mu}$ and the covariance matrix C using the different estimators described in the previous section: the naive sample estimator, the rolling window estimator, the EWMA estimator with $\lambda = 0.94$, and the Ledoit-Wolf shrinkage estimator. The James-Stein estimator was used for the mean vector. For each combination of estimators, the optimal Markowitz portfolio weights were computed by solving the quadratic program described in Section 1, either via the closed-form analytical solution derived using Lagrange multipliers, or via numerical quadratic programming solvers for the constrained long-only case. A natural difficulty in evaluating portfolio performance is that asset prices tend to rise on average over long horizons, meaning that almost any invested portfolio will show positive cumulative returns simply due to the general upward trend of equity markets. A rising portfolio value is therefore not, by itself, evidence that the optimization model is adding value. To rigorously assess the effectiveness of each method, we therefore compare its performance against two reference benchmarks:

- **Equally-weighted portfolio:** each asset receives a weight $w_i = 1/n$, regardless of its return or risk. This naive allocation serves as a baseline representing the performance achievable with no optimization at all, and is known to be surprisingly competitive in practice.
- **S&P 500 index:** the performance of the broad U.S. equity market, representing the opportunity cost of simply holding a passive index fund rather than running an active mean-variance optimizer.

The comparison is conducted via **backtesting**.

To evaluate a portfolio method honestly, we must separate what is *known at decision time* from what happens *after* the decision. This is the core logic of our backtesting framework.

1) Walk-forward principle. At each rebalancing date t , the model is calibrated only on a past estimation window $[t - W, t)$, where W is the training length. The computed weights are then held over the next holding window $[t, t + H)$, where H is the rebalancing step. This rolling train-then-test cycle is repeated through time.

2) No look-ahead bias. Future returns are never used to estimate parameters or choose weights. In other words, the strategy at time t only depends on information available up to $t - 1$.

3) Dynamic portfolio value. If $r_{p,t}$ is the realized portfolio return at date t , the net asset value evolves as:

$$\text{NAV}_t = \text{NAV}_{t-1}(1 + r_{p,t}).$$

This recursion gives the full performance path, not only an average return.

4) Transaction costs and turnover. Rebalancing is not free. Let $\mathbf{w}_t$ be new weights and $\mathbf{w}_{t-1}$ previous weights. Turnover is measured by

$$\tau_t = \sum_{i=1}^n |w_{i,t} - w_{i,t-1}|,$$

and a proportional cost is deducted from returns at each rebalance. This prevents overly optimistic results for highly reactive strategies.

5) Relative evaluation. The strategy is compared to a simple benchmark (equal-weight) under the same market period. This allows us to interpret whether complexity actually adds value in terms of return, risk, drawdown, and stability.

Overall, the framework answers a practical question: *If we had followed this rule in real time, with realistic frictions, would performance remain robust out-of-sample?*

Reproducibility and Interactive Dashboard:

All figures, models, and backtesting results presented in this report are fully reproducible. The complete Python codebase is available on the project’s GitHub [repository](#), where each figure can be regenerated by running the corresponding script with adjustable parameters (universe of assets, estimation window, optimization method, backtesting period).

In addition, an interactive dashboard built with **Streamlit** is available in the `benchmark_demo` folder of the repository. To launch it, install the required dependencies listed in `requirements.txt` and run the following command in the terminal:

```
pip install -r requirements.txt
streamlit run Cockpit.py
```

The dashboard allows the user to interactively select the asset universe, the estimation method, the optimization constraints, and the backtesting period, and displays in real time the resulting efficient frontier, optimal weights, and out-of-sample performance track record compared to the two benchmarks.

3.1 Empirical Results on Real S&P 500 Data

Whereas the two illustrative figures above use offline synthetic prices for reproducibility, this section evaluates the *entire* pipeline on **real** market data. We keep every modelling hypothesis of the experiment settings unchanged, daily simple returns, a sector-stratified universe, $\hat{\mu}$ = sample mean, $\hat{\Sigma}$ = EWMA ($\lambda = 0.94$), the max-Sharpe objective solved with **CVXPY/CLARABEL** under long-only and a per-asset cap $w_{\max} = 0.15$, an estimation window $W = 252$ days, a holding period $H = 21$ days, transaction costs of 10 bps, and a risk-free rate $r_f = 4\%$. Only the data source changes from synthetic to real, and we add a third, fully passive benchmark: the **S&P 500 index** (`^GSPC`), which represents the opportunity cost of simply buying the market. After requiring a complete 2019–2025 price history, 32 assets are retained; the out-of-sample track record therefore runs from early 2020 (after the first estimation window) and includes the COVID crash and the 2022 rate-tightening drawdown.

Table 1 reports the out-of-sample metrics. Three observations stand out. First, the optimizer does *not* earn a higher return than the naive $1/N$ allocation (CAGR 12.8% vs 12.5%); what it improves is **risk**: lower volatility (21.2% vs 22.8%) and a shallower maximum drawdown (−34.6% vs −40.0%), which lifts the Sharpe ratio from 0.37 to 0.42. This is precisely the channel through which a better Σ adds value. Second, even so, both active and naive equity portfolios remain *below* the capitalization-weighted S&P 500 (Sharpe 0.45), which rode a handful of mega-cap stocks over the period: the index is genuinely hard to beat. Third, turnover is low (3% per rebalance) and no optimization fallback was triggered, confirming the numerical robustness of the convex solver on this universe.

Strategy	CAGR %	Vol %	Sharpe	Max DD %	Turnover
Markowitz max-Sharpe	12.80	21.16	0.42	−34.62	0.03
Equal-weight $1/N$	12.45	22.83	0.37	−40.04	—
S&P 500 index	13.50	20.94	0.45	−33.92	—

Table 1: Out-of-sample performance on real S&P 500 data (2020–2025, 32 assets), strategy vs the two benchmarks. Sharpe ratios are net of the 4% risk-free rate and of 10 bps transaction costs.

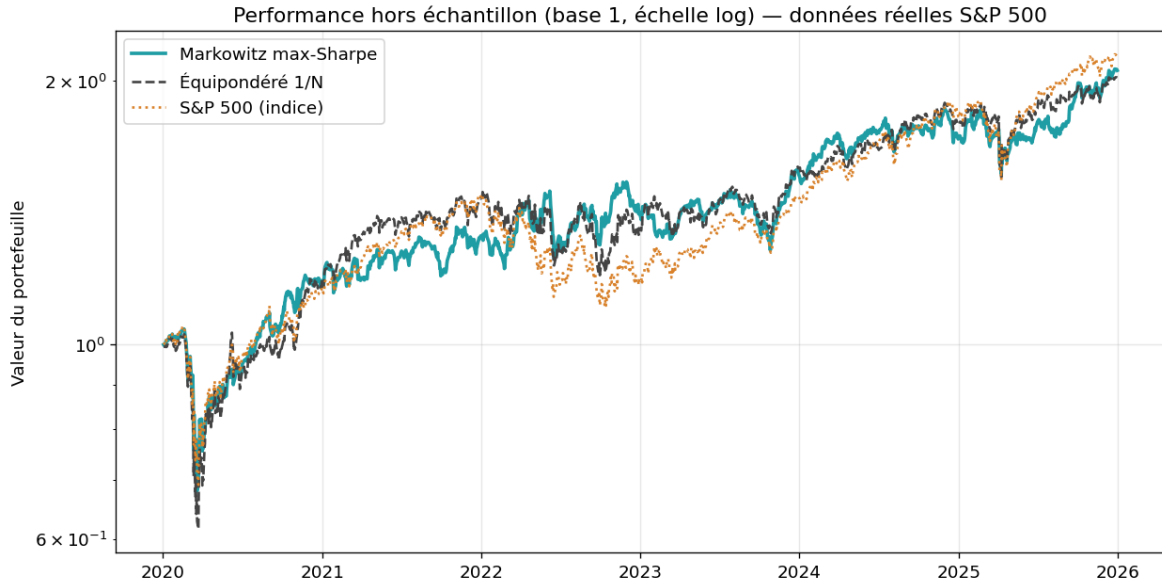


Figure 10: Out-of-sample net asset value (base 1, log scale): Markowitz max-Sharpe vs equal-weight $1/N$ vs the S&P 500 index, on real data. The three paths track closely; the strategy's edge is in risk control, not in raw return.



Figure 11: Drawdown of the strategy through time. The deepest losses coincide with the COVID crash (2020) and the 2022 tightening.

To make the link between drawdowns and *exogenous* market shocks explicit, Figure 12 overlays the two largest real stress episodes of the period (the February–April 2020 COVID crash and the 2022 monetary tightening) on the same value curves. The strategy and both benchmarks lose together during these episodes, illustrating the well-known fact that diversification erodes exactly when correlations spike toward one.

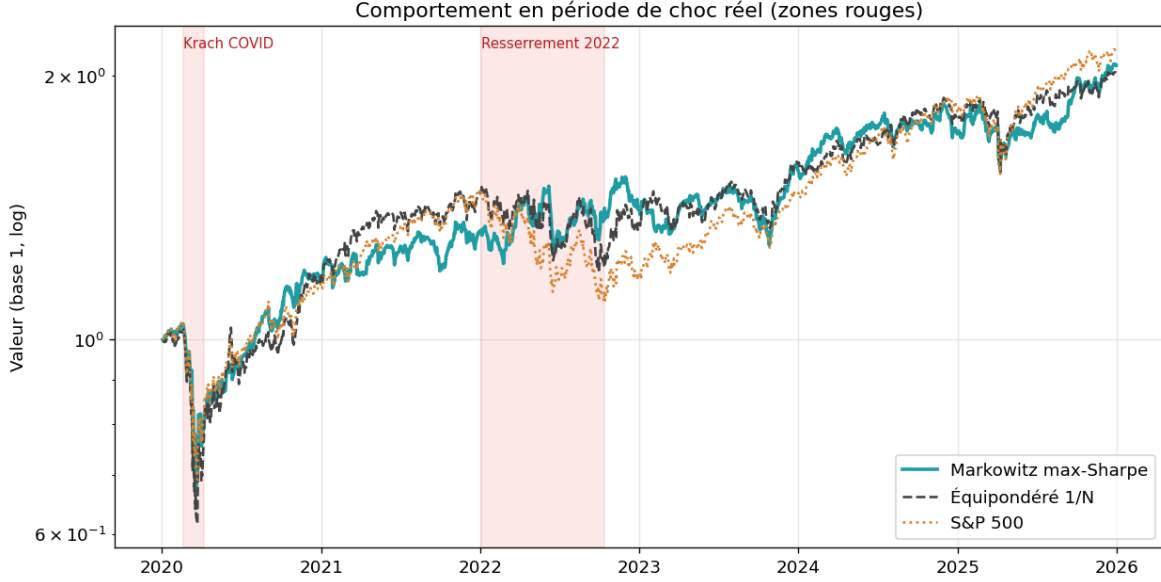


Figure 12: Value paths with the two real shock windows shaded in red. Drawdowns concentrate inside the stress periods, where cross-asset correlations rise and diversification benefits collapse.

Ex-post Validation: do $\hat{\mu}$ and $\hat{\Sigma}$ hold?

A backtest tells us whether a strategy performed well, but not *why*. To diagnose the source of performance, we validate the two statistical ingredients **out-of-sample**: at each rebalancing date we compare the quantity estimated on the past window to the quantity actually realized over the following holding window.

The expected return $\hat{\mu}$ does not hold. The natural metric is the *information coefficient* (IC), the Spearman rank correlation between predicted and realized returns across assets. Figure 13 shows that the IC is statistically indistinguishable from zero for *every* estimator we tried, sample mean, EWMA, James-Stein, rolling mean, and a PCA factor model, with the sample mean at $IC \approx 0.01$. The point cloud of predicted versus realized annualized returns (Figure 14) confirms this visually: it is an isotropic blob with a slightly *negative* correlation (-0.06), nowhere near the $y = x$ line of a good forecast. In plain terms, past average returns carry essentially no information about future average returns, the classic finding of DeMiguel et al. (2009), and the reason μ -dependent objectives struggle to beat $1/N$.

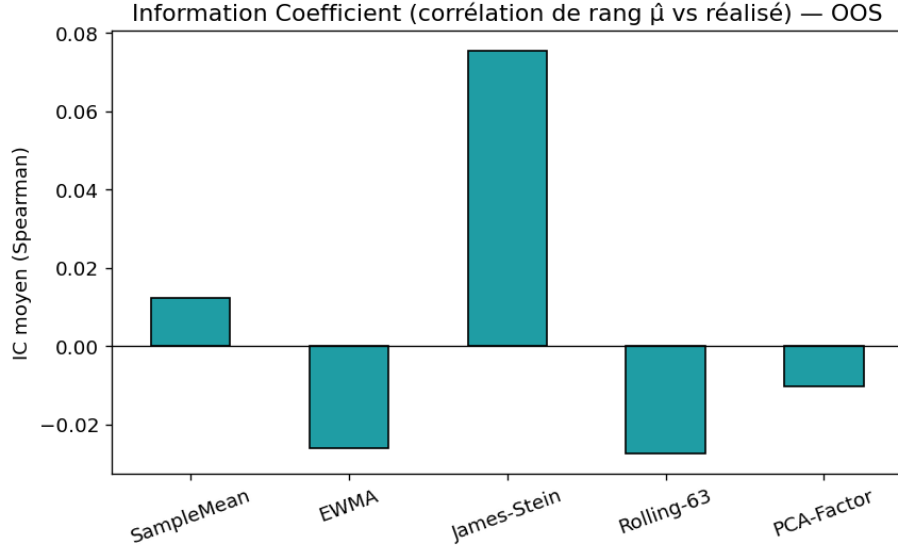


Figure 13: Out-of-sample information coefficient (Spearman rank correlation of $\hat{\mu}$ vs realized returns) for five mean estimators. All are ≈ 0 : no estimator predicts the cross-section of returns.

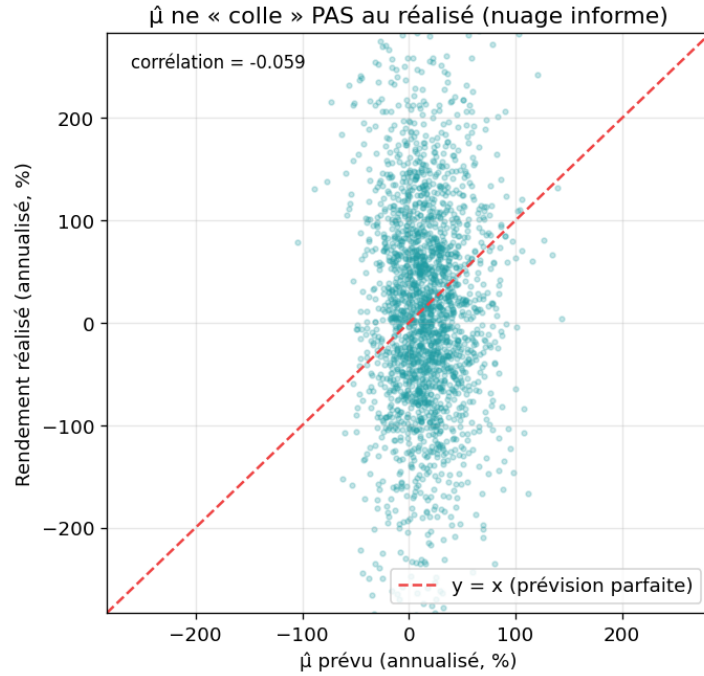


Figure 14: Predicted $\hat{\mu}$ vs realized return (annualized, all assets and windows). The cloud is informationless: $\hat{\mu}$ does *not* stick to the realized return.

The covariance $\hat{\Sigma}$ does hold. The same exercise applied to risk tells the opposite story. Figure 15 ranks the covariance estimators by out-of-sample Gaussian log-likelihood (higher is better) and by absolute volatility-forecast error (lower is better); the regularized estimators (Ledoit-Wolf, EWMA) dominate the raw sample covariance. Crucially, the scatter of predicted versus realized annualized volatility (Figure 16) lines up along the $y = x$ diagonal, with a clearly positive correlation (0.54): the volatility forecast by $\hat{\Sigma}$ is a genuine predictor of realized

volatility. This asymmetry, $\hat{\Sigma}$ is **reliable** while $\hat{\mu}$ is **not**, is the central empirical message of the project and explains why the value of optimization lies in risk control rather than return prediction.

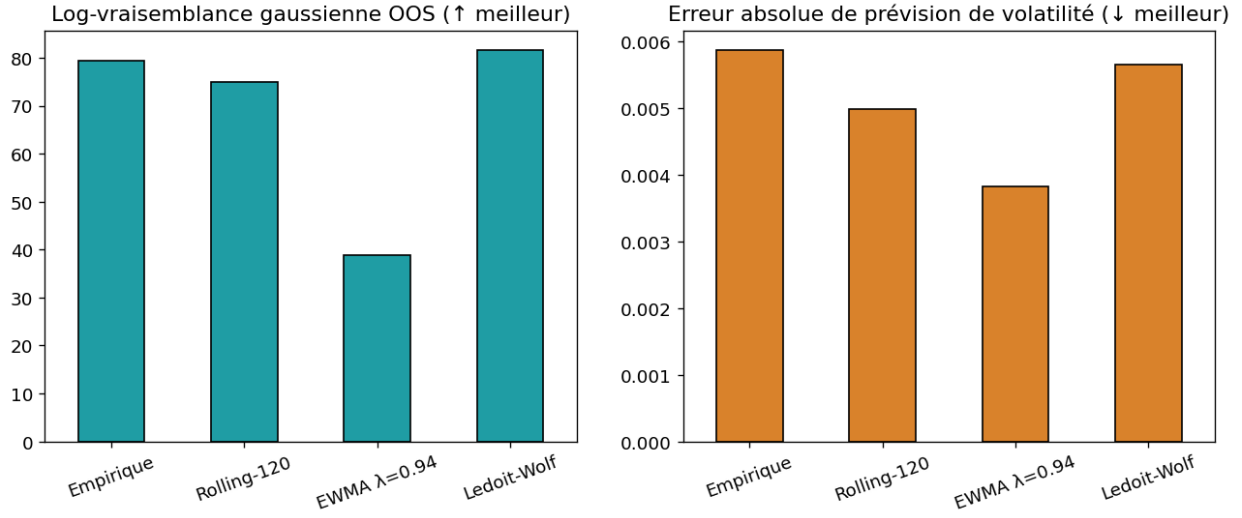


Figure 15: Out-of-sample statistical quality of four covariance estimators: Gaussian log-likelihood (left, $\uparrow$) and volatility-forecast error (right, $\downarrow$). Shrinkage/EWMA estimators beat the raw sample covariance.

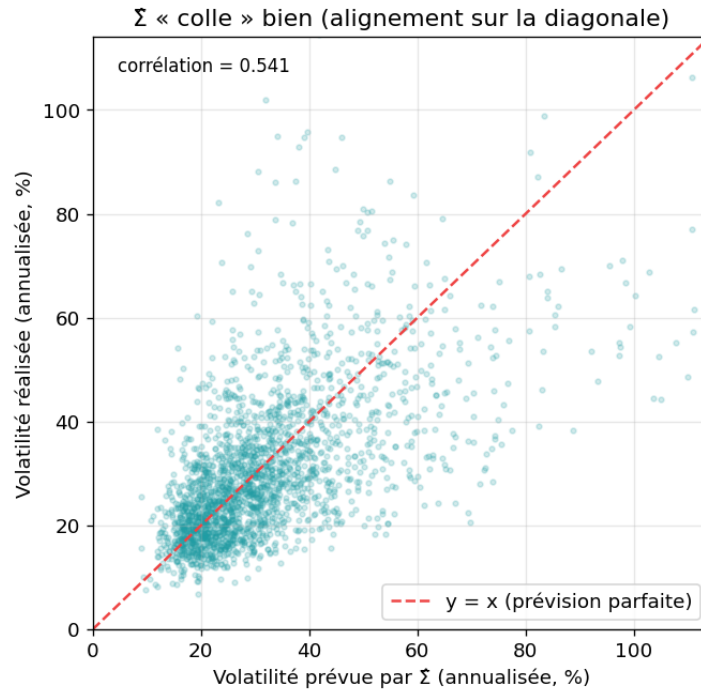


Figure 16: Volatility predicted by $\hat{\Sigma}$ vs realized volatility (annualized, all assets and windows). Points align on the diagonal: $\hat{\Sigma}$ sticks to reality.

Limitations and Assumptions

While methodologically careful, our framework rests on several assumptions discussed throughout this report. They are summarized below to define the scope of our conclusions:

- **Gaussian, i.i.d. returns.** Discussed in Part 2.
- **Estimation error dominates, especially on μ .** Discussed in Part 2.
- **Static covariance regime within each window.** Even the EWMA and shrinkage estimators assume a locally stationary covariance over the estimation window; abrupt regime changes (e.g. the COVID crash) are detected only with a lag.
- **Simplified frictions.** Transaction costs are modelled as a proportional charge on turnover; market impact, bid-ask spreads that vary with liquidity, financing costs and taxes are ignored. We also restrict ourselves to long-only, fully-invested, single-currency equity portfolios, without leverage or short positions.
- **Effective universe size.** After requiring a complete price history over the test period, the cleaned universe is smaller than the nominal target; combined with the per-asset weight cap, this limits the room the optimizer has to deviate from equal weighting.

These limitations point directly to the natural extensions discussed in the conclusion: robust (M-)estimation of μ and Σ , regime-aware covariance prediction, and tail-risk objectives.

Conclusion

This project re-implemented the Markowitz mean-variance framework end to end, from the closed-form and KKT theory of Part 1, through the statistical estimation of its two ingredients in Part 2, to a fully out-of-sample, cost-aware backtest on real S&P 500 data in Part 3. Three conclusions emerge, and they are consistent across the analytical derivations, the simulations and the empirical evidence.

(i) **The expected-return vector μ is, for practical purposes, unpredictable.** No estimator, however sophisticated (shrinkage, EWMA, factor models), produces an information coefficient distinguishable from zero out-of-sample, and this holds across estimation windows.

(ii) **The covariance matrix Σ , by contrast, is genuinely predictable:** its volatility forecasts line up with realized volatility, and regularized estimators (Ledoit-Wolf, EWMA) improve both conditioning and forecast quality.

(iii) **Because mean-variance optimization is most sensitive to errors in μ ,** the dimension we estimate worst, the theoretical optimum is fragile, the equally-weighted $1/N$ portfolio is hard to beat, and the value an optimizer can realistically add lies in *risk control* (a better Σ , sensible constraints) rather than in *return forecasting*. On our data the optimized portfolio did beat $1/N$ on a risk-adjusted basis, by lowering volatility and drawdown rather than by raising return, yet still trailed the capitalization-weighted index.

The most promising directions for future work follow directly from these findings and from the limitations above: replacing the sample mean and covariance with *robust M-estimators* that bound the influence of outliers, adopting *regime-aware* covariance predictors of the kind recently proposed by Johansson et al. (2023), and complementing variance with explicit *tail-risk* objectives (CVaR). All of these can be slotted into the modular pipeline and the interactive dashboard developed here without altering the evaluation protocol.

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